Q.20 Compare regenerative and rheostatic braking systems.

Q.21 Explain briefly the working of a pantograph.

Q.22 What are the uses of battery systems in electric trains?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the classification of electric traction systems and explain any two with neat sketches.

Q.24 Define and draw the speed-time curve for suburban services. Explain the significance of each section of the curve.

Q.25 Explain in detail the working and importance of the overhead catenary wire and pantograph system in electric traction.

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Roll No. ....

5th Sem / Electrical

Subject : Electrical Traction System

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is a key environmental benefit of electric traction?

a) Higher fuel usage b) Noise pollution

c) Lower emission d) High thermal loss

Q.2 Which traction system is most commonly used in Indian Railways for long-distance trains?

a) DC traction

b) Diesel traction

c) 25 kV single-phase AC traction

d) Three-phase AC traction

Q.3 Coasting is defined as:

a) Running at full power

b) Braking with full force

c) Running without power supply

d) Changing direction

Q.4 Which of the following is an electrical braking method?

a) Regenerative braking

b) Air braking

c) Hydraulic braking

d) Manual braking

Q.5 Why is the DC series motor preferred for traction?

a) High speed regulation

b) Low starting torque

c) High starting torque

d) Low voltage operation

Q.6 Pantograph slides over which part of the system to collect current?

a) Rails b) Battery bank

c) Overhead wire d) Control box

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Name two systems of electric traction.

Q.8 Define schedule speed.

Q.9 Mention any two essential features of a traction motor.

Q.10 State one advantage of thyristor-based control in traction.

Q.11 Write any one advantage of electric braking.

Q.12 Name one type of auxiliary equipment used in electric traction.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain advantages and disadvantages of DC traction system.

Q.14 What is a composite traction system? Give one real-life example.

Q.15 Differentiate between schedule speed and average speed.

Q.16 How do curves and gradients affect train performance.

Q.17 Describe the working of a DC traction motor with the help of a diagram.

Q.18 What are the conditions required for successful regenerative braking?

Q.19 Write short notes on :

i) Tap changer

ii) Chopper Control